Collagen Biosynthesis

Learning Objectives

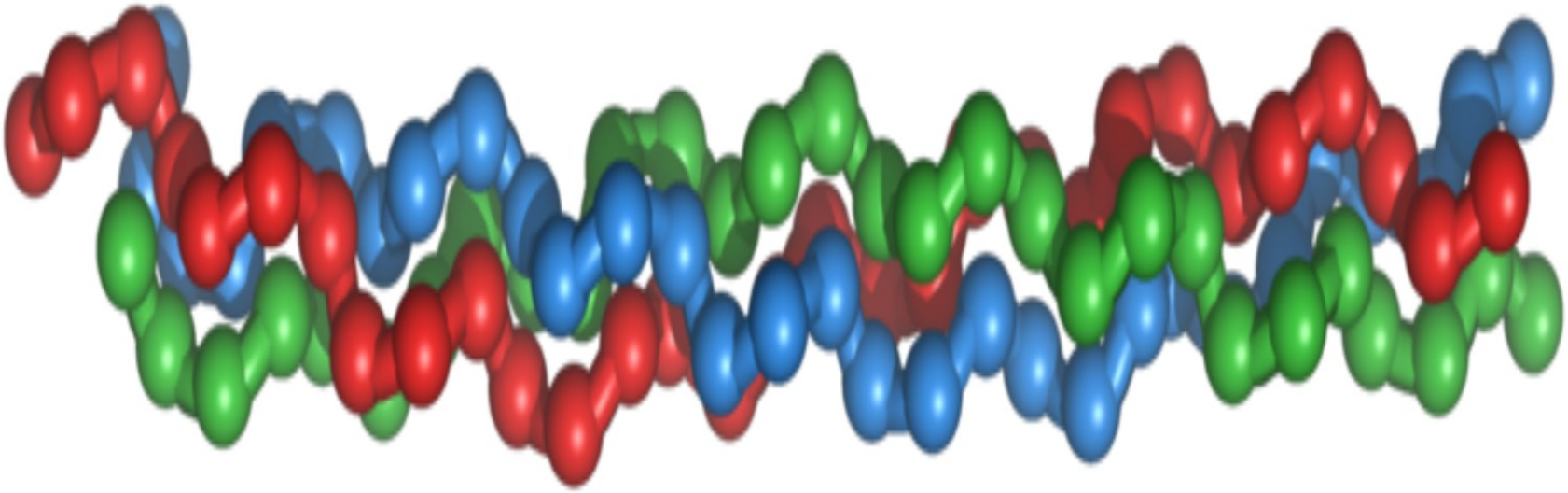
At the end of this session student will be able to understand;

1. What is collagen?
2. Its Biosynthesis and types.
3. Diseases related to its defective synthesis.

Collagen

- Collagen - name given to a family of structurally related proteins that form strong insoluble fibers.
- Synthesized on ribosome.
- Major component of most connective tissues, constitute approximately 25% of protein of mammals and it is the most abundant protein in the animal world.

- It provides extracellular framework for all animals and exists in every animal tissues.
- At least 25 distinct types of collagen made up of over 30 distinct polypeptide chains have been identified in human tissues.
- Collagen consists of three polypeptide chains.
- Different types of collagen are found in different locations in the body.



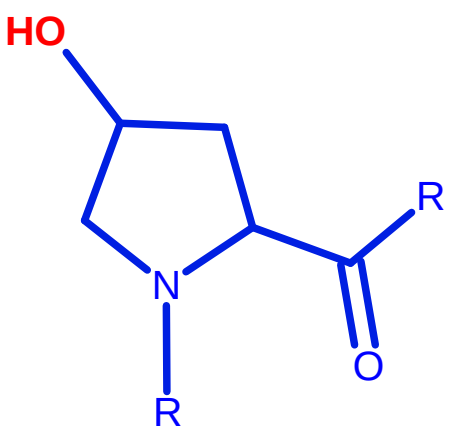
Collagen - overview

- Collagen – are post-translationally modified by hydroxylation and glycosylation and on transport through RER and Golgi.
- Each molecule is composed of a left-handed helix.
- Three such helices are coiled to form a right-handed superhelix

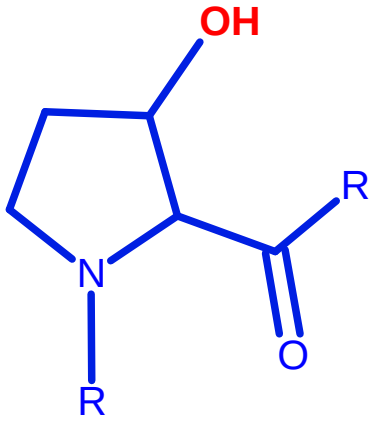
Collagen - Composition

- One third (30%) of residues are Glycine
- Other (~ 30) quarter of residues are Proline or Hydroxyproline (HyPro)
- 5-hydroxylysine (HyLys) also occurs; a site for glycosylation
- Hydroxylation of Pro, Lys is a post-translational modification by the action of proline hydroxylase and lysine hydroxylase.
- These Fe^{2+} requires vitamin C as a reactant
- The sequence of collagen bears long stretches of

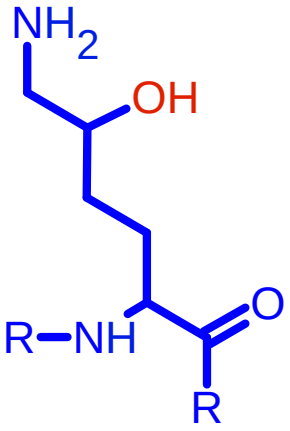
  Gly--Pro/HyPro-X repeats



4-HyPro



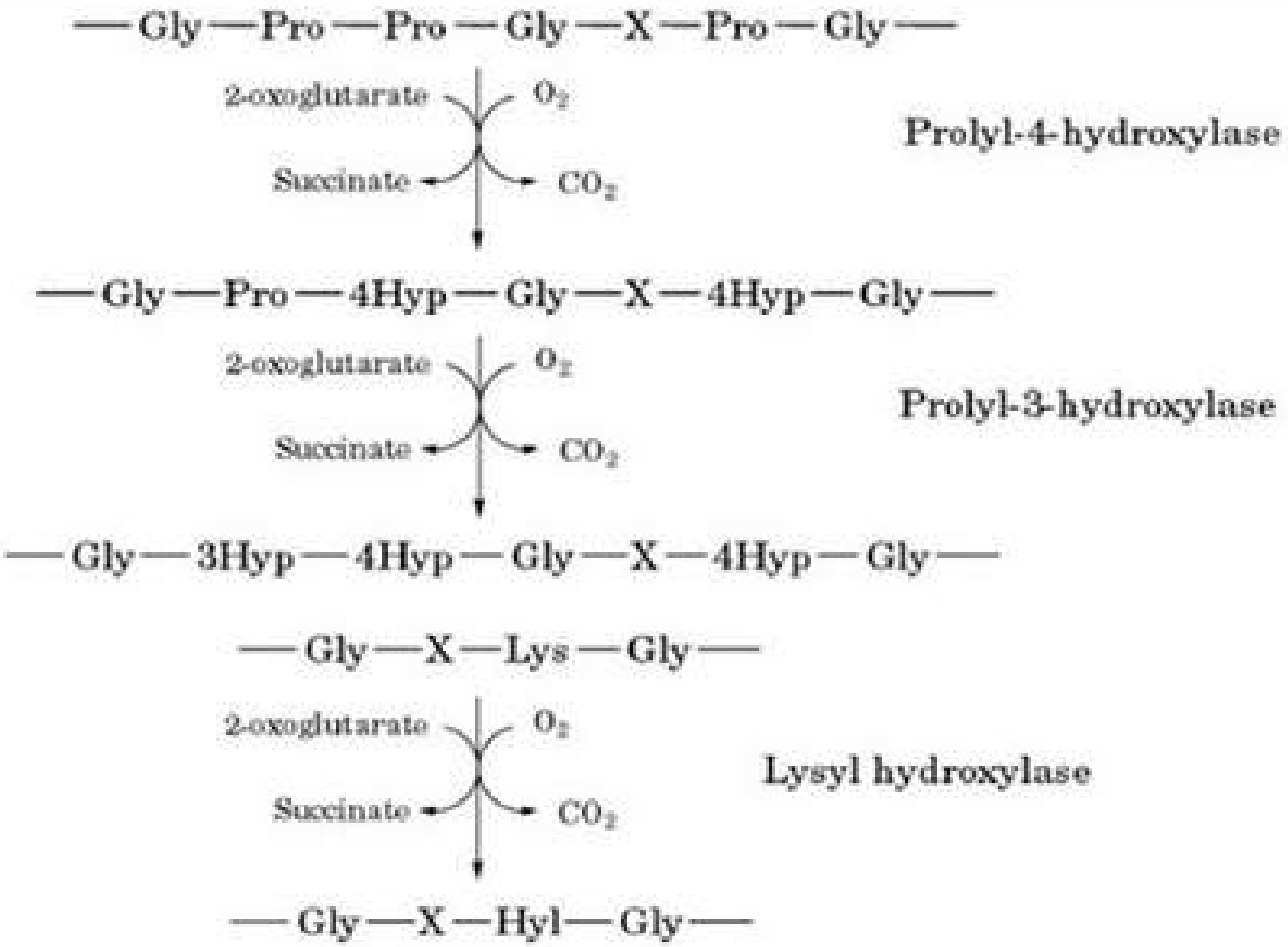
3-HyPro



5-HyLys

Collagen - It is found in all forms of connective tissue.

Collagen itself is an insoluble protein because of extensive cross- linking.


— Gly — Pro — Pro — Gly — X — Pro — Gly —
2-oxoglutarate
O_2
Succinate
CO_2
Prolyl-4-hydroxylase
— Gly — Pro — 4Hyp — Gly — X — 4Hyp — Gly —
2-oxoglutarate
O_2
Succinate
CO_2
Prolyl-3-hydroxylase
— Gly — 3Hyp — 4Hyp — Gly — X — 4Hyp — Gly —
— Gly — X — Lys — Gly —
2-oxoglutarate
O_2
Succinate
CO_2
Lysyl hydroxylase
— Gly — X — Hyl — Gly —

Structure of Collagen (continued)

- The structural unit of collagen is a tropocollagen, a supercoil made up of 3 helices, with a molecular mass of ~285 kdal.
- Each collagen helix consists of ~ 1000 amino acid residues. The helix is left-handed. **It is not an a-helix.**
- The helix contains 3 amino acids per turn, with a pitch of 0.94 nm.

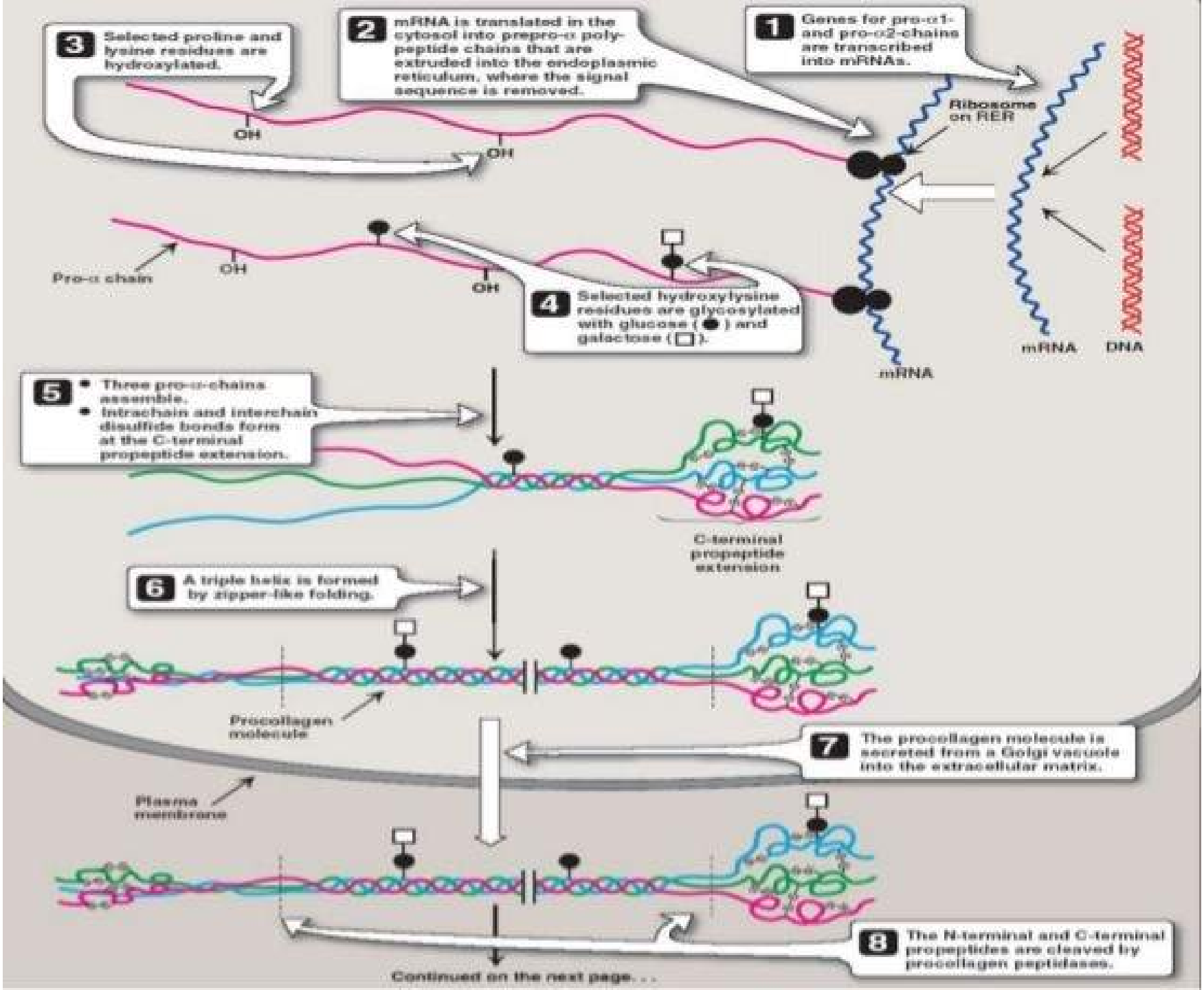
Structure of Collagen (continued)

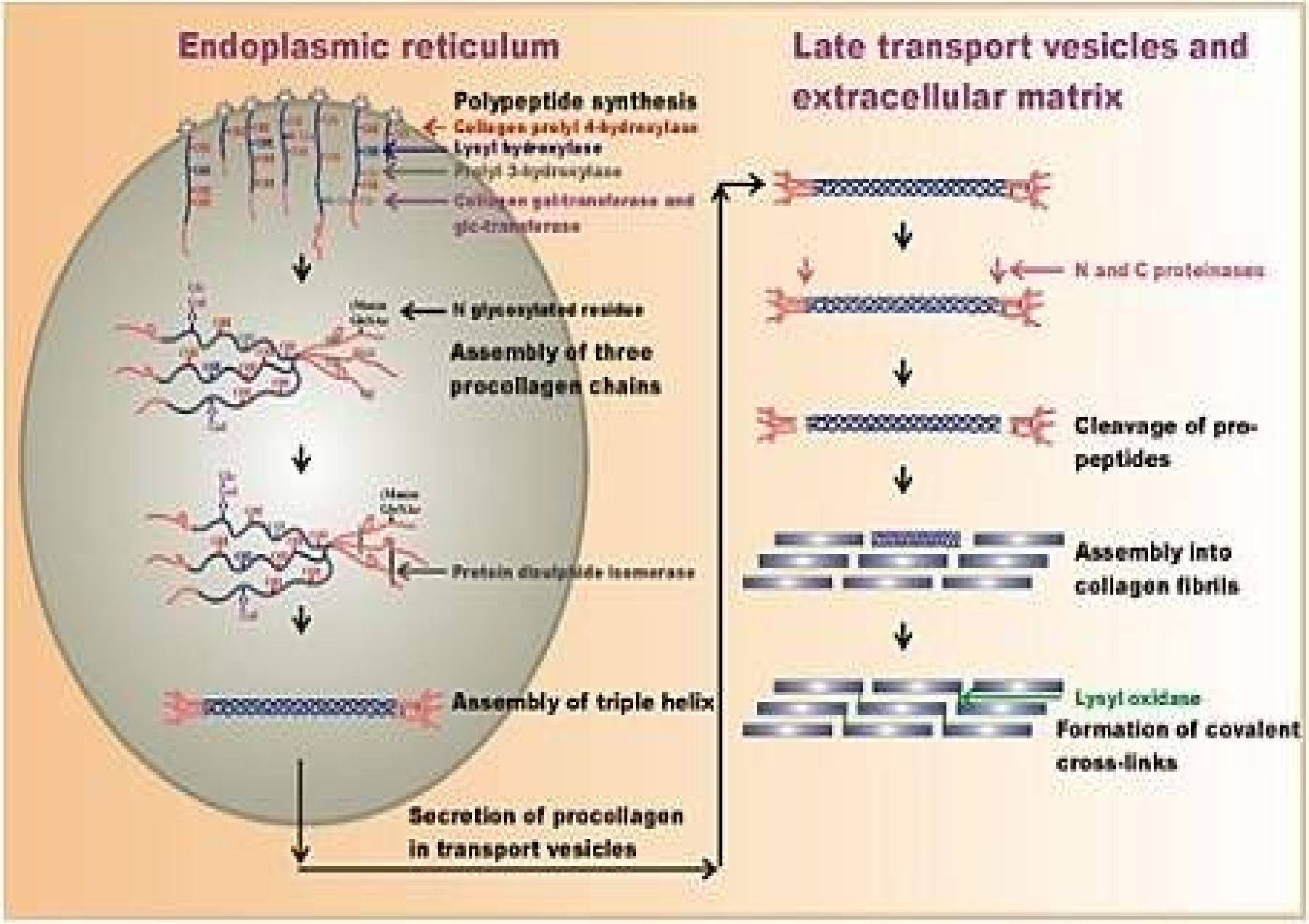
- Each unit of tropocollagen is about 1.5 nm wide and 300 nm long. Bundles of these 3-stranded supercoils can be seen as collagen fibres in the electron microscope.
- Mature collagen has extensive covalent crosslinkages between individual collagen molecules.

Steps in collagen synthesis

- Biosynthesis of collagen is divided into 3 steps;

1. Intracellular steps.
2. Secretion.
3. Extra-cellular steps.


1 Genes for pro-α1- and pro-α2-chains are transcribed into mRNAs.
2 mRNA is translated in the cytosol into prepro-α polypeptide chains that are extruded into the endoplasmic reticulum, where the signal sequence is removed.
3 Selected proline and lysine residues are hydroxylated.
Ribosome on RER
OH
OH
Pro-α chain
OH
OH
4 Selected hydroxylysine residues are glycosylated with glucose (●) and galactose (□).
mRNA
mRNA
DNA
5 • Three pro-α-chains assemble.
• Intrachain and interchain disulfide bonds form at the C-terminal propeptide extension.
C-terminal propeptide extension
6 A triple helix is formed by zipper-like folding.
Procollagen molecule
7 The procollagen molecule is secreted from a Golgi vacuole into the extracellular matrix.
Plasma membrane
8 The N-terminal and C-terminal propeptides are cleaved by procollagen peptidases.
Continued on the next page. . .


Endoplasmic reticulum
Late transport vesicles and extracellular matrix
Polypeptide synthesis
Collagen prolyl 4-hydroxylase
Lysyl hydroxylase
Prolyl 3-hydroxylase
Collagen galctransferase and glc-transferase
N glycosylated residue
Assembly of three procollagen chains
Protein disulphide isomerase
Assembly of triple helix
Secretion of procollagen in transport vesicles
N and C proteinases
Cleavage of pro-peptides
Assembly into collagen fibrils
Lysyl oxidase
Formation of covalent cross-links

Synthesis of collagen

Chains



Procollagen



Tropocollagen

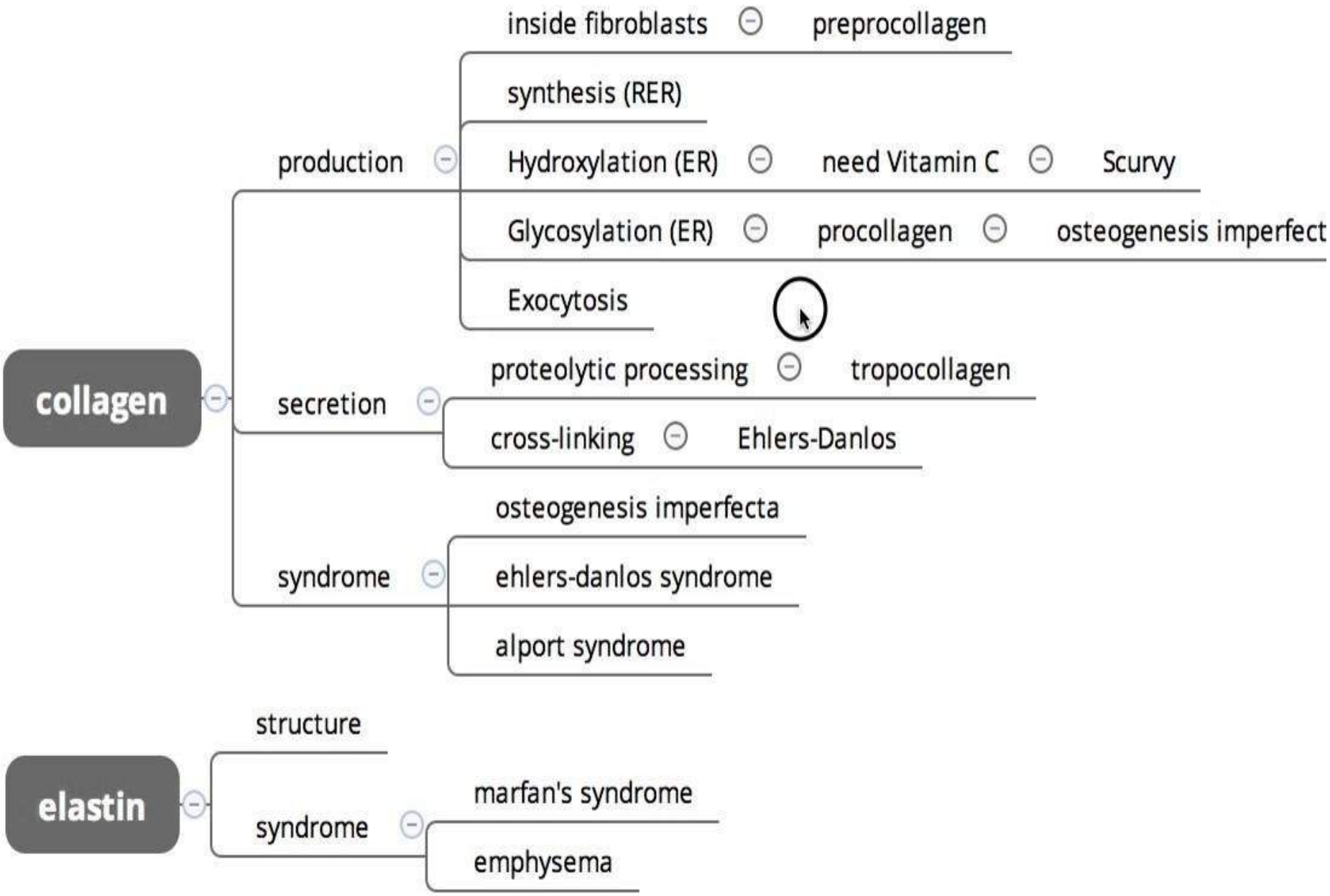


Fibril

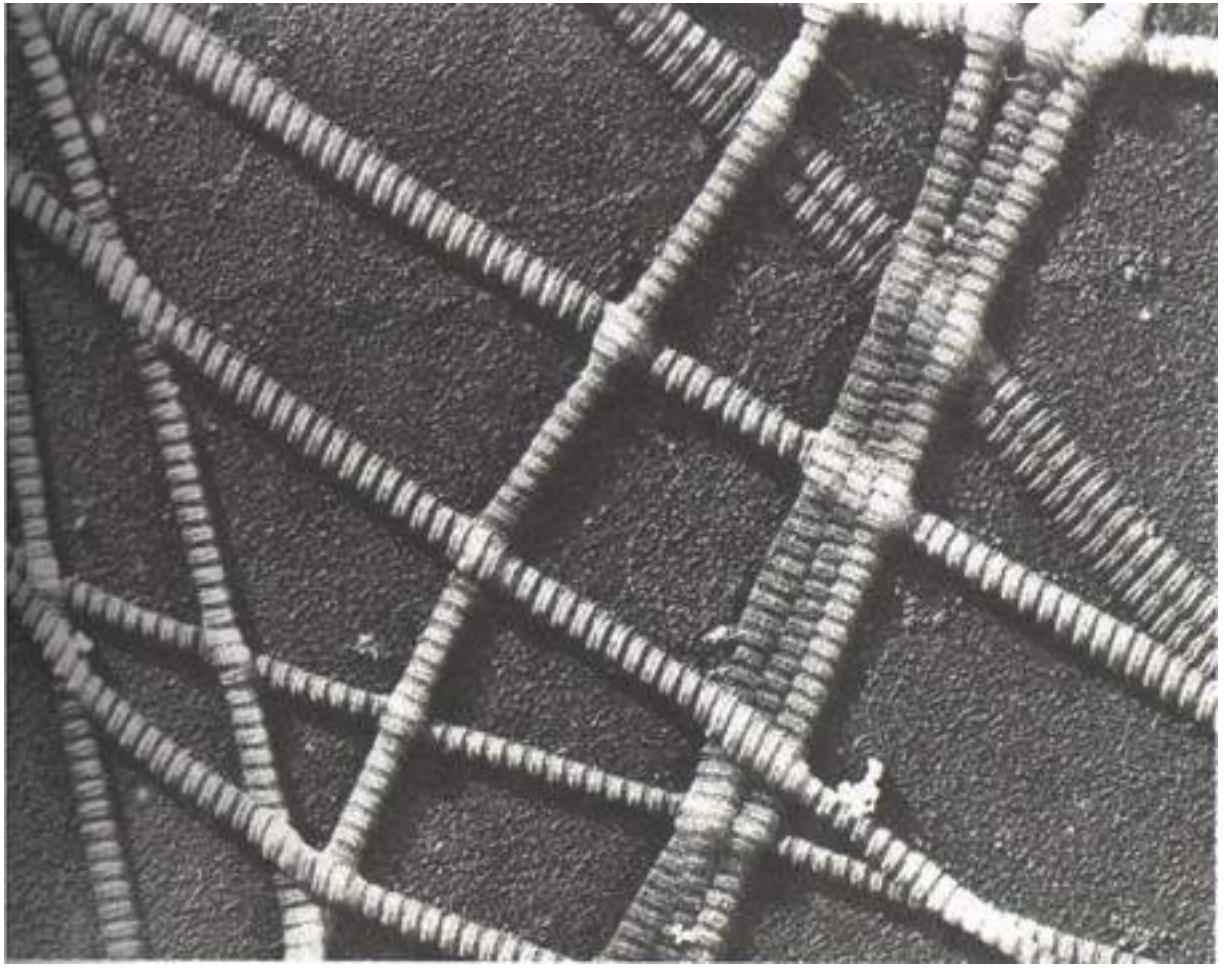


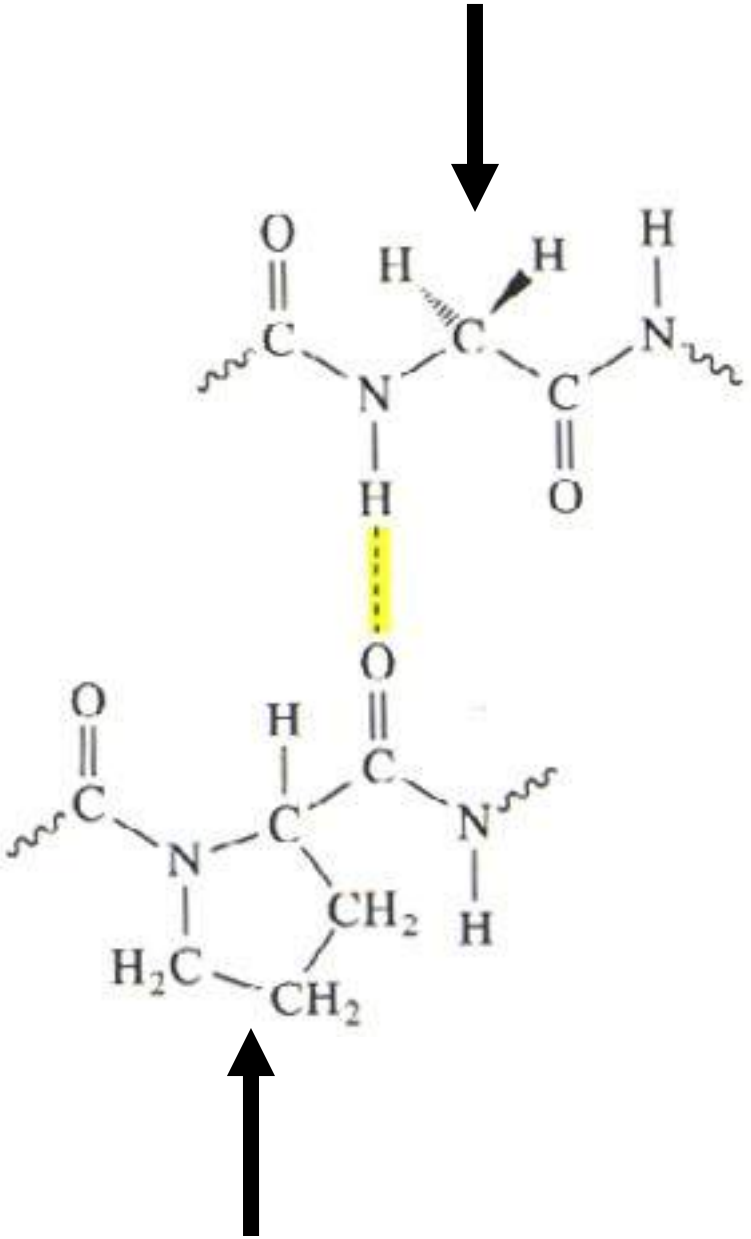
Fibre




collagen
production
inside fibroblasts
preprocollagen
synthesis (RER)
Hydroxylation (ER)
need Vitamin C
Scurvy
Glycosylation (ER)
procollagen
osteogenesis imperfect
Exocytosis
secretion
proteolytic processing
tropocollagen
cross-linking
Ehlers-Danlos
syndrome
osteogenesis imperfecta
ehlers-danlos syndrome
alport syndrome
elastin
structure
syndrome
marfan's syndrome
emphysema

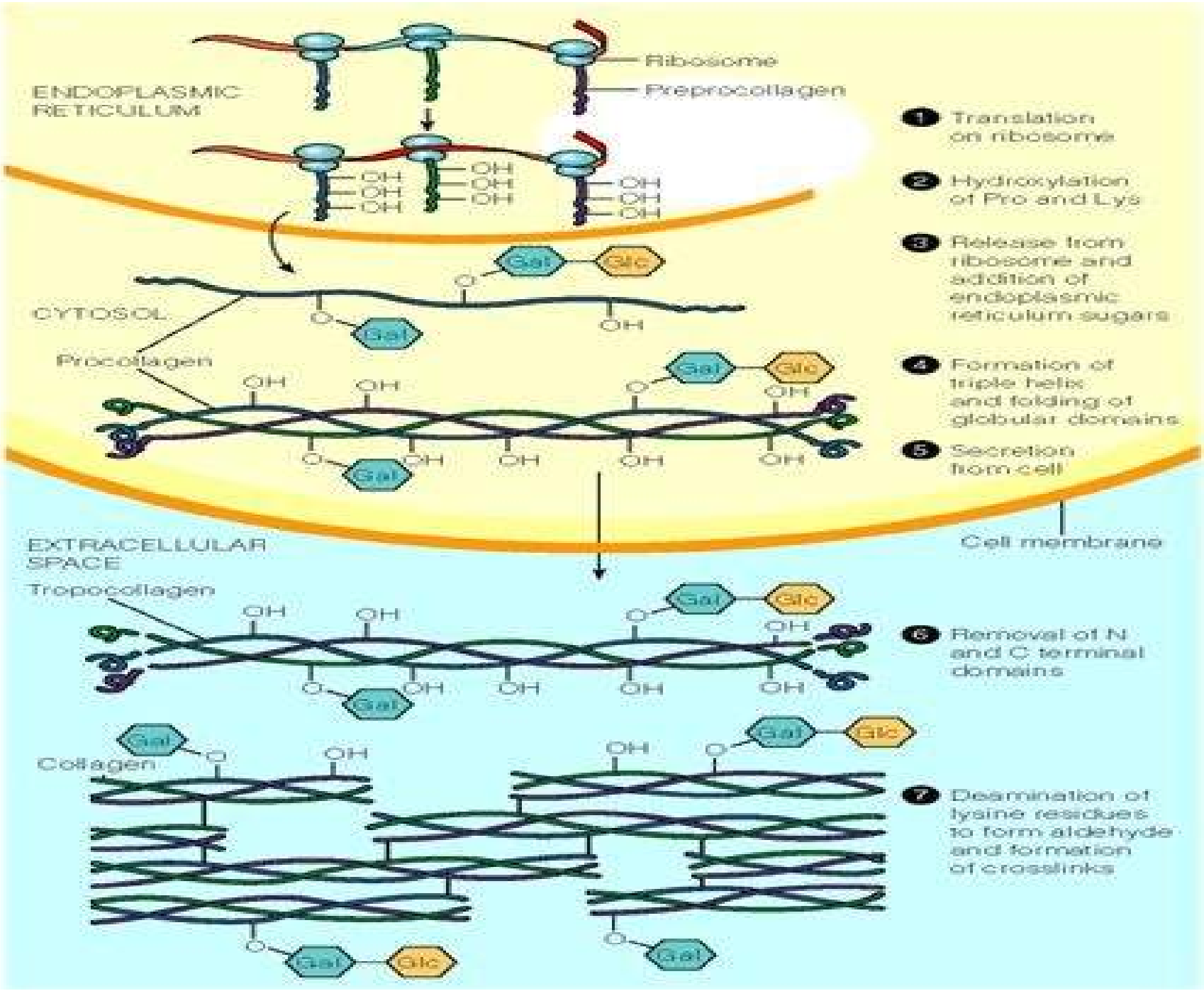
An electron micrograph of collagen from skin

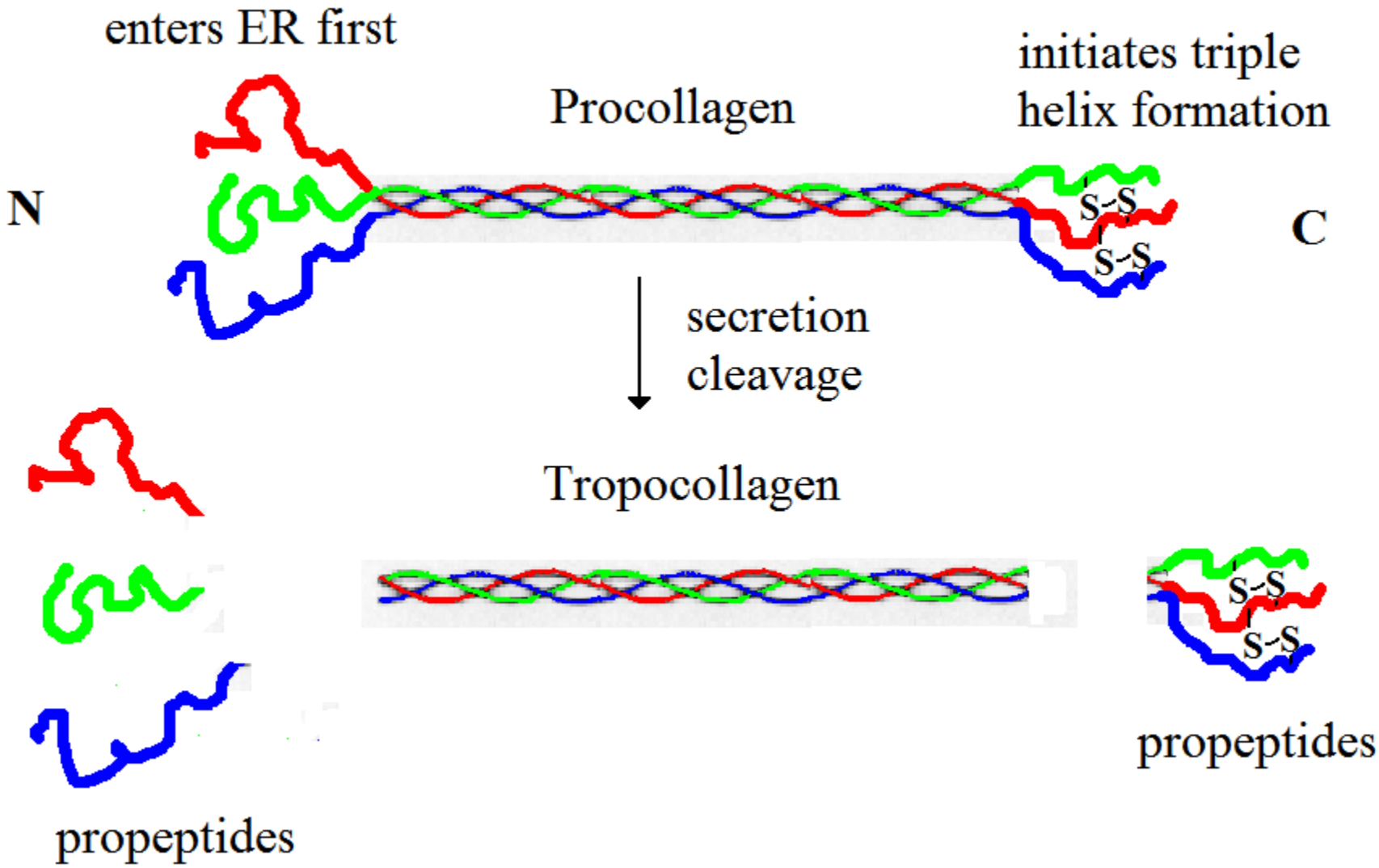




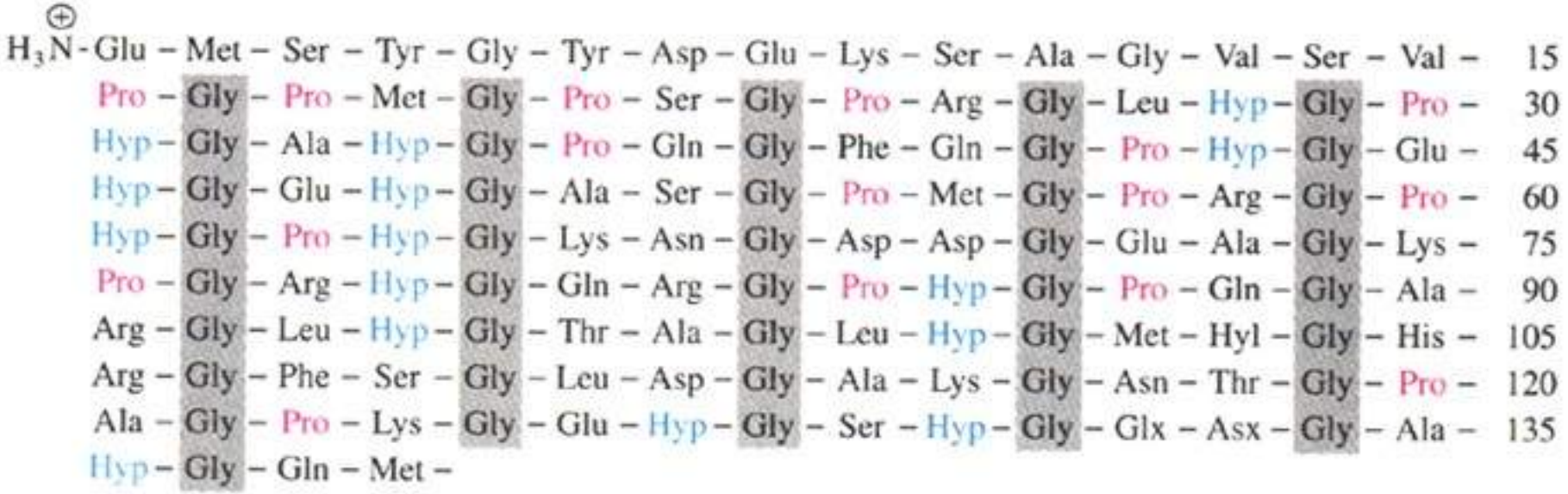
Inter-chain H-bonding between a glycine residue

And a proline residue of a different chain


ENDOPLASMIC RETICULUM
Ribosome
Preprocollagen
OH
Gal
Glc
CYTOSOL
Procollagen
1 Translation on ribosome
2 Hydroxylation of Pro and Lys
3 Release from ribosome and addition of endoplasmic reticulum sugars
4 Formation of triple helix and folding of globular domains
5 Secretion from cell
Cell membrane
EXTRACELLULAR SPACE
Tropocollagen
6 Removal of N and C terminal domains
Collagen
7 Deamination of lysine residues to form aldehyde and formation of crosslinks


enters ER first
Procollagen
initiates triple
helix formation
N
S-S
S-S
C
secretion
cleavage
Tropocollagen
S-S
S-S
propeptides
propeptides

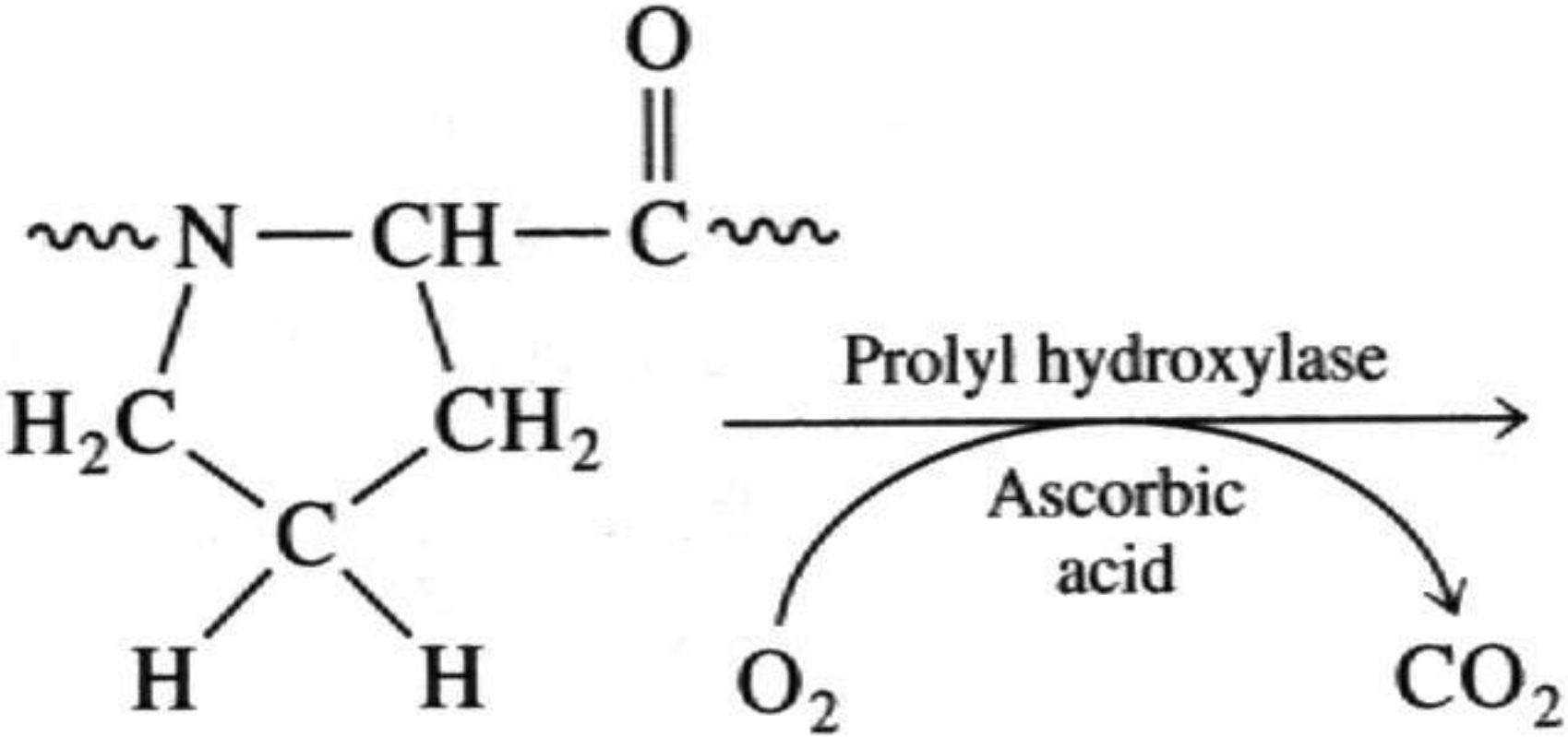
Amino acid sequence of collagen



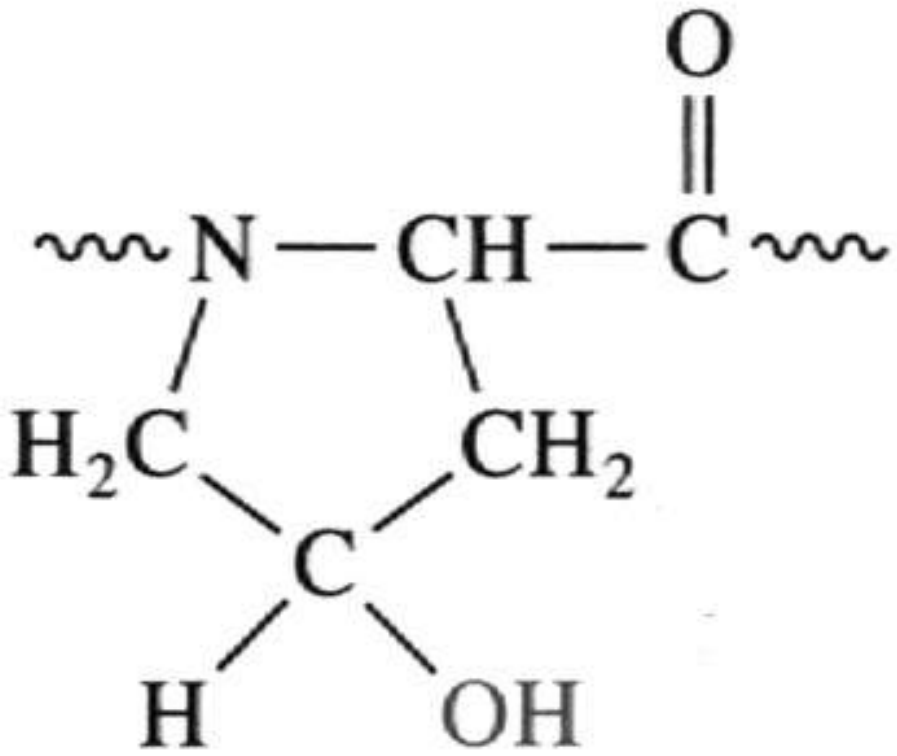
Defective Hydroxylation Is One Of The Biochemical Lesions in Scurvy

The importance of the hydroxylation of collagen becomes evident in *scurvy*.

In the vitamin C deficiency disease scurvy, collagen does not form correctly due to the inability to hydroxylate Pro and Lys.



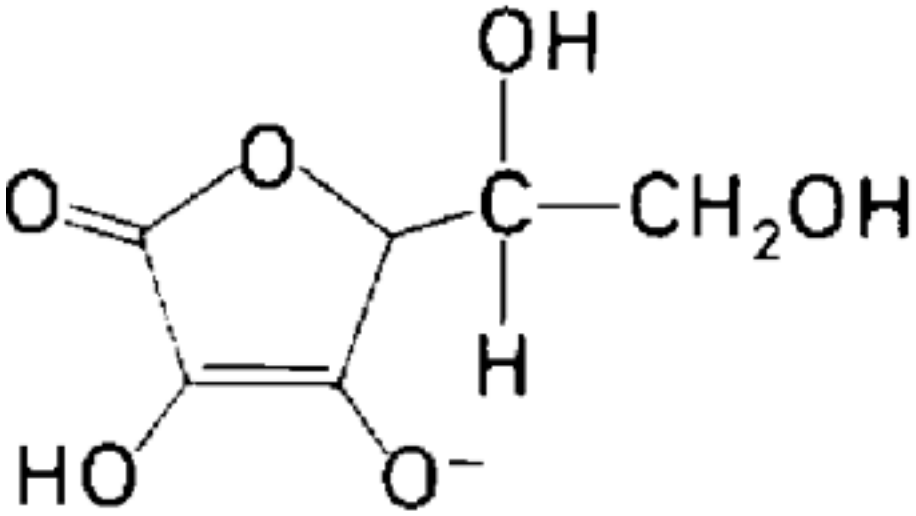
Proline residue



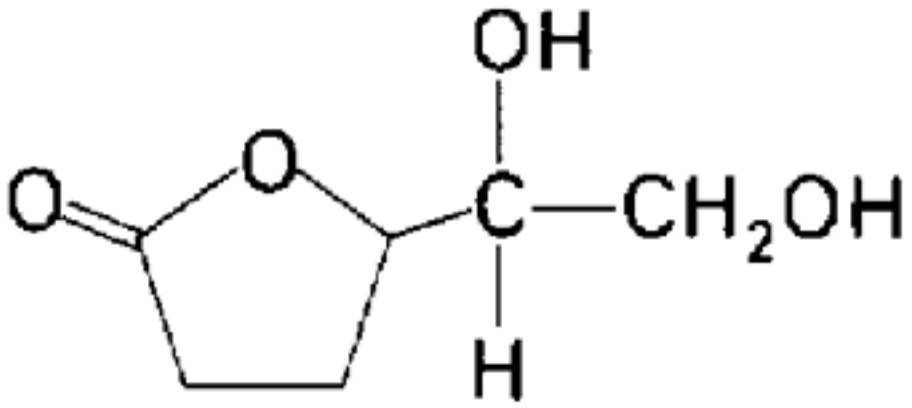
4-Hydroxyproline residue

The presence of hyp residues greatly increases the potential for H-bonding between chains.

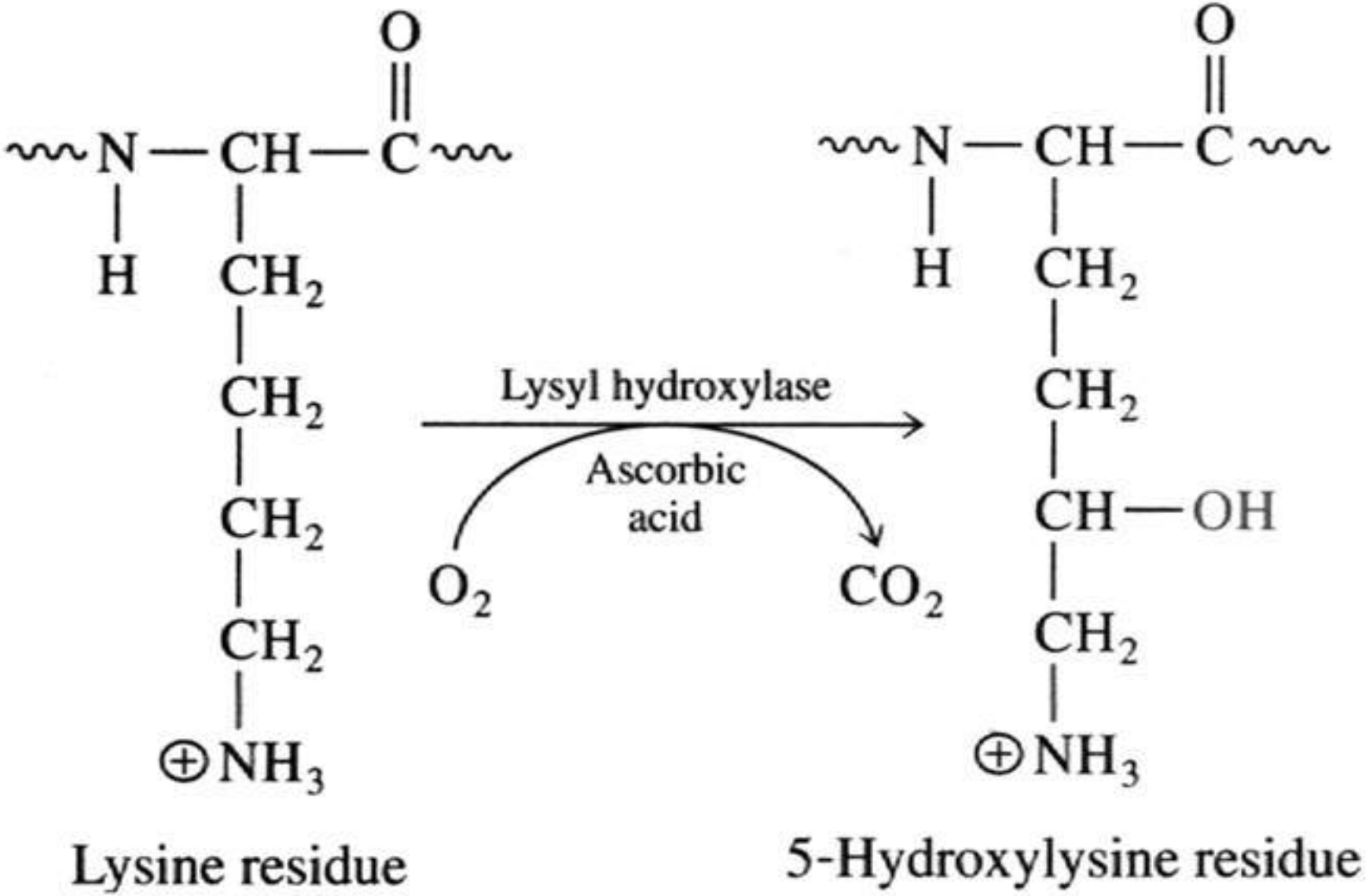
Hyp and pro make up 25% of the residues of collagen



Ascorbate



Dehydroascorbic acid



Some Hydroxy lysine residues are covalently bonded to carbohydrates making collagen a glycoprotein

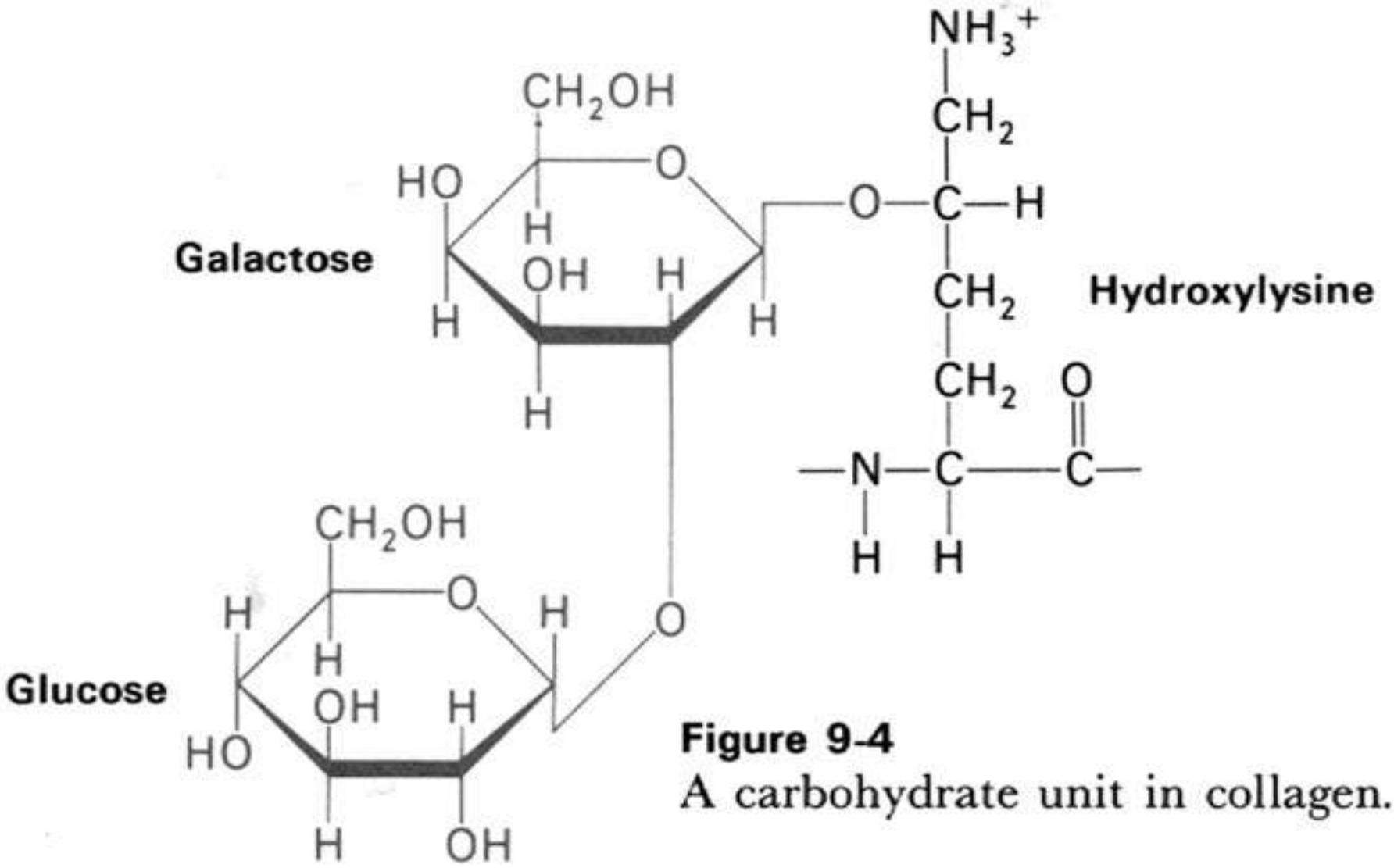
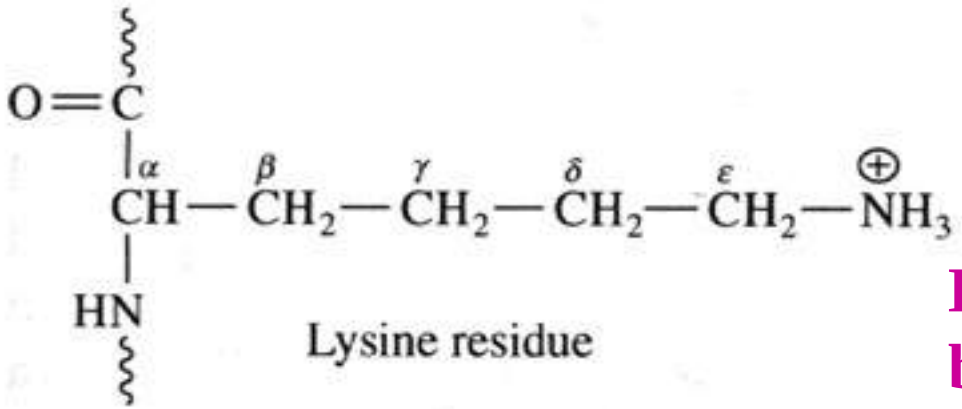
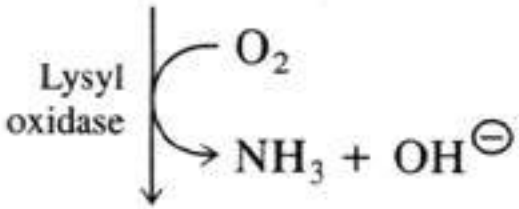


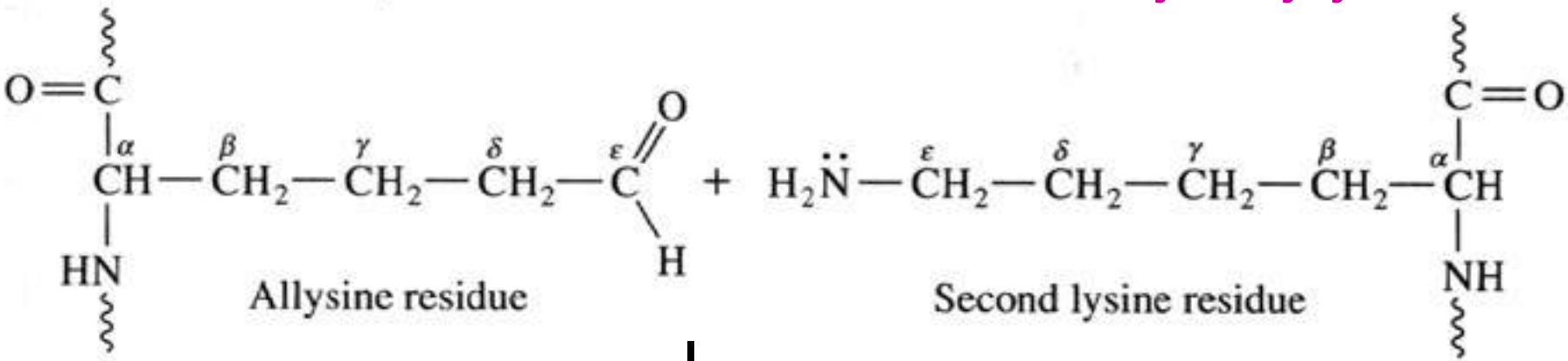
Figure 9-4
A carbohydrate unit in collagen.



Lysine residue



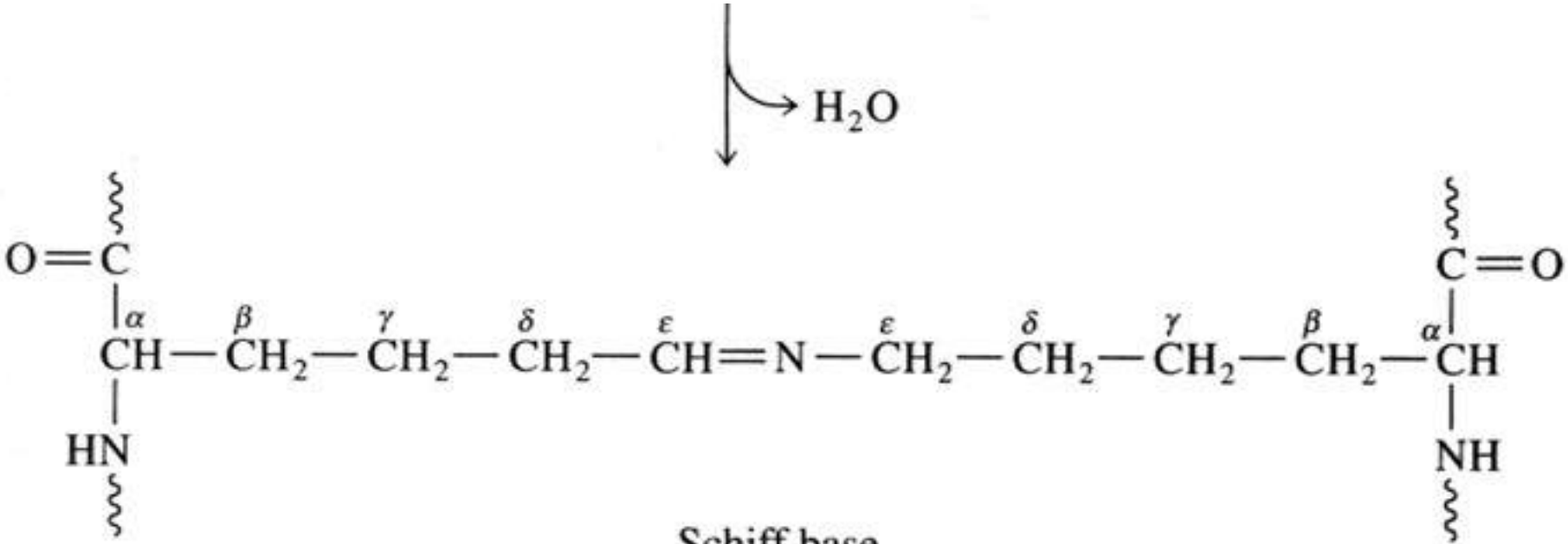
Lathyrism : abnormalities of bones, joints and blood vessels. *Lathyrus odoratus* seeds contain β-aminoproprionitrile, which blocks the enzyme lysyl oxidase



Allysine residue

Second lysine residue





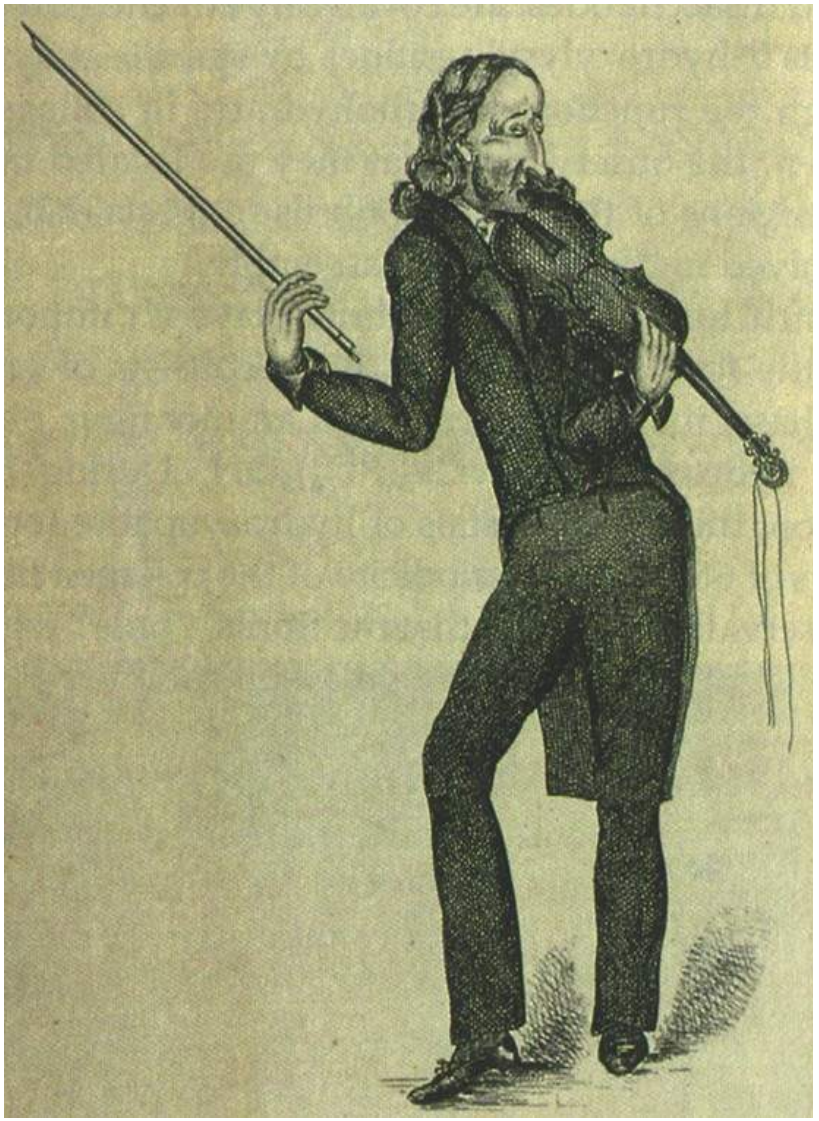
Schiff base

Genetic disorders of collagen are numerous :

One such group of 10 different collagen deficiency diseases is the Ehlers-Danlos (E-D) syndrome.

The “India-rubber man” of circus fame had an E-D syndrome.

Osteogenesis imperfecta : abnormal (fragile) bone formation in human babies



Paganini may also have suffered from an E-D condition known as Marfan’s syndrome.

Ehlers-Danlos Syndrome

Hyperplasticity of the skin



Osteogenesis imperfecta (brittle bone disease)

- Type II :
- Most severe form, frequently lethal at or shortly after birth, often due to respiratory problems.
- In recent years, some people with Type II have lived into young adulthood.
- Numerous fractures and severe bone deformity; small stature with underdeveloped lungs.
- Collagen is improperly formed.